

Experiment 1: 8-Queen Problem

Aim

To implement the 8-Queen problem using Artificial Intelligence techniques so that eight queens are placed on a chessboard without attacking each other.

Objective

To solve the 8-Queen problem using the Backtracking algorithm.

Algorithm

1. Start from the first row.
2. Place a queen in the first safe column.
3. Move to the next row.
4. Check whether the current position is safe.
5. If safe, place the queen.
6. If no safe position exists, backtrack to the previous row.
7. Continue until all eight queens are placed.
8. Display the solution.

Flowchart

Start

|



Initialize Chessboard

|

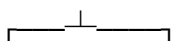


Place Queen in Row

|



Is Position Safe?



Yes No

|

|

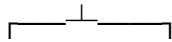


Next Row Next Column

|



All Queens Placed?



Yes No

|

|



Display Backtrack

Solution

|



End

Python Code

N = 8

```
def print_solution(board):
```

```
    for i in range(N):
```

```
        print(board[i])
```

```
def is_safe(board, row, col):
```

```
    for i in range(col):
```

```
        if board[row][i] == 1:
```

```
            return False
```

```
i = row
j = col
while i >= 0 and j >= 0:
    if board[i][j] == 1:
        return False
    i -= 1
    j -= 1
```

```
i = row
j = col
while i < N and j >= 0:
    if board[i][j] == 1:
        return False
    i += 1
    j -= 1
```

```
return True
```

```
def solve(board, col):
    if col >= N:
        return True

    for i in range(N):
        if is_safe(board, i, col):
            board[i][col] = 1

            if solve(board, col + 1):
                return True
```

```
board[i][col] = 0
```

```
return False
```

```
board = [[0] * N for _ in range(N)]
```

```
if solve(board, 0):
```

```
    print_solution(board)
```

```
else:
```

```
    print("No Solution")
```

Sample Output

```
[1, 0, 0, 0, 0, 0, 0, 0]
```

```
[0, 0, 0, 0, 1, 0, 0, 0]
```

```
[0, 0, 0, 0, 0, 0, 0, 1]
```

```
[0, 0, 0, 0, 0, 1, 0, 0]
```

```
[0, 0, 1, 0, 0, 0, 0, 0]
```

```
[0, 0, 0, 0, 0, 0, 1, 0]
```

```
[0, 1, 0, 0, 0, 0, 0, 0]
```

```
[0, 0, 0, 1, 0, 0, 0, 0]
```

Result

Thus, the 8-Queen problem was successfully solved using the Backtracking algorithm.